



Jet Engine Performance

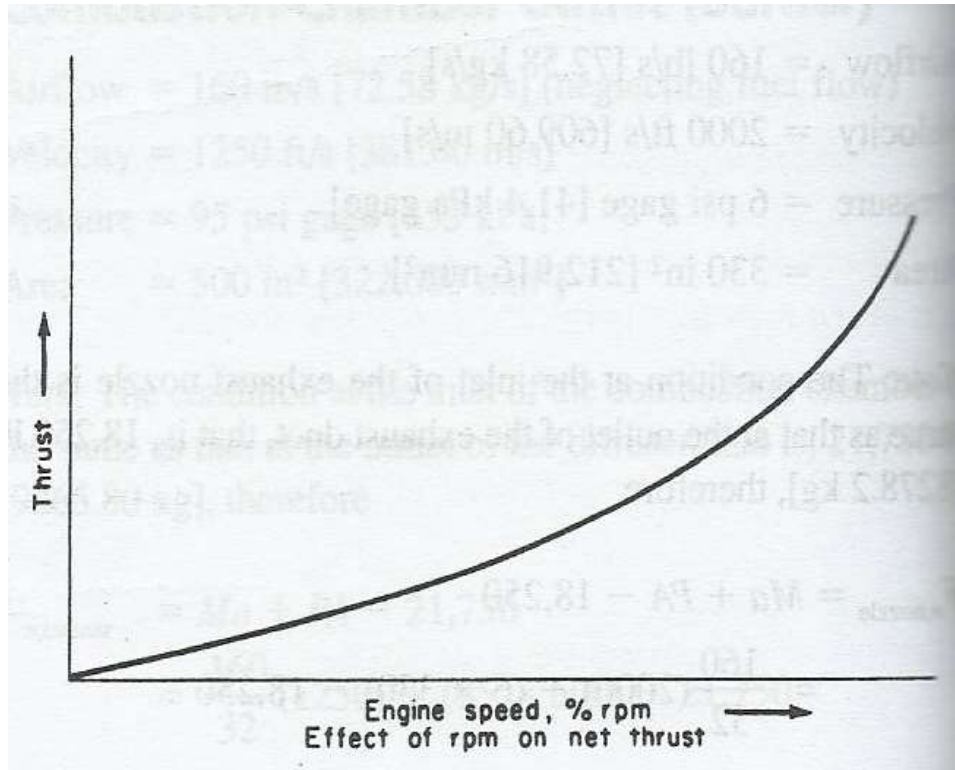
OPERATING PARAMETERS OR FACTORS AFFECTING THRUST

Various parameters which affect gas turbine engines are

1. Engine RPM
2. Speed
3. Temperature
4. Pressure
5. Density

EFFECT OF RPM

- RPM of an engine greatly affects the thrust of the engine
- RPM is directly proportional to the thrust

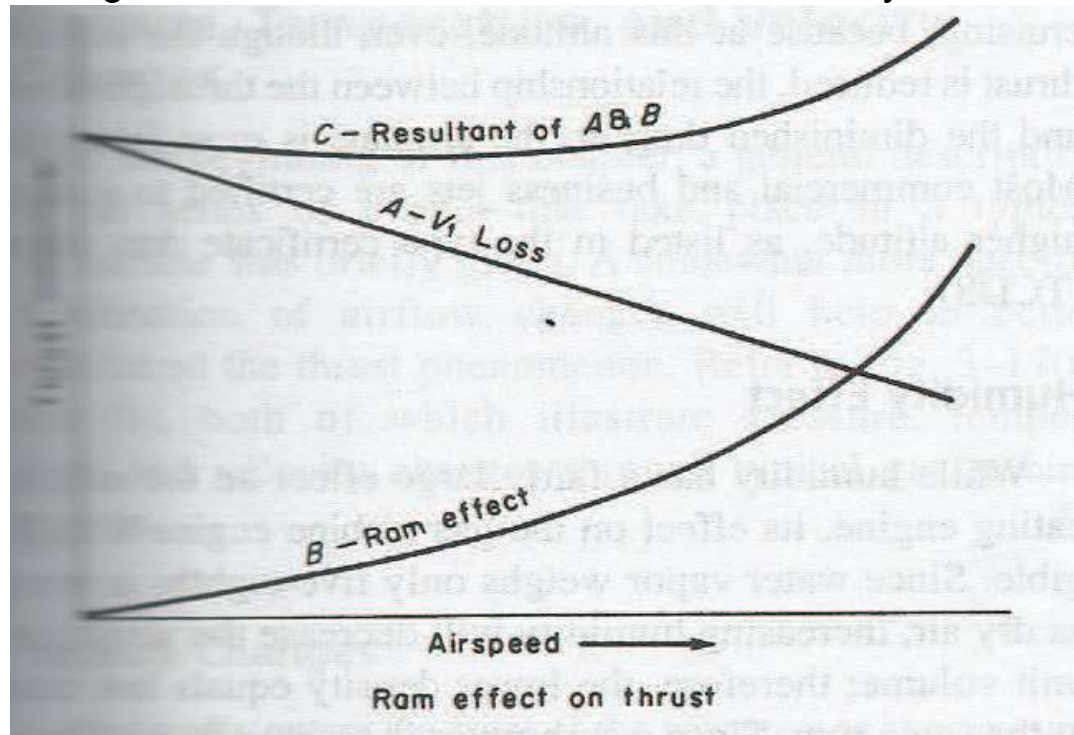


EFFECT OF SPEED

➤ W.K.T

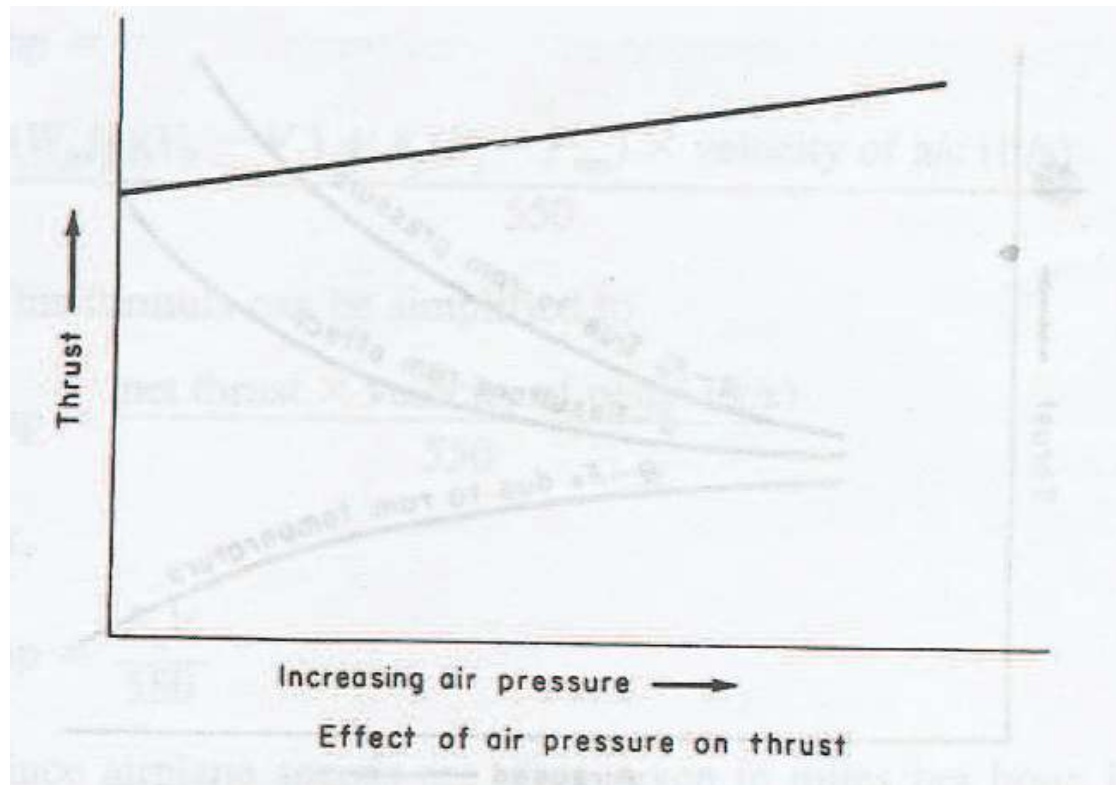
$$F = m \times a$$
$$F = \frac{W a (V_2 - v_1)}{g}$$

- As the forward velocity increases, the difference in term ($v_2 - v_1$) decreases and hence the thrust.
- But in reality, as the forward speed increases the amount of mass packed into the engine increases and hence the thrust.
- By combining the above two effects the thrust actually increases.



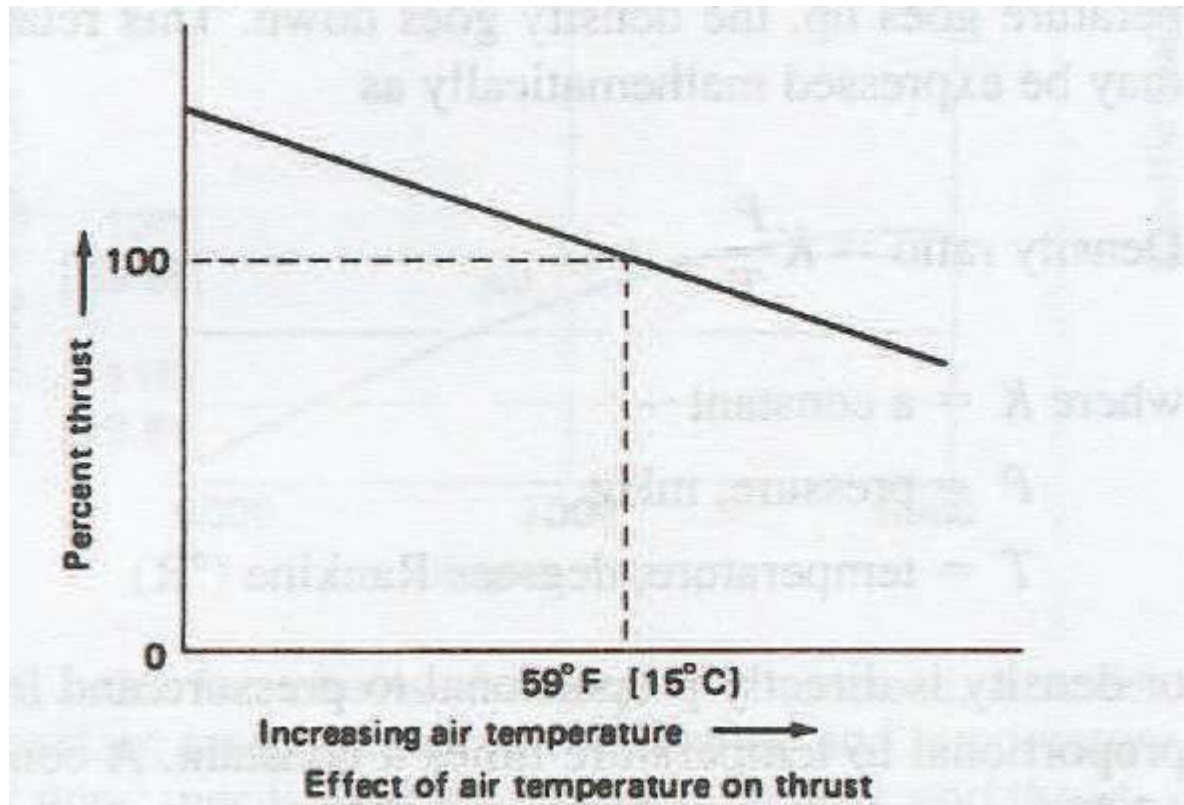
EFFECT OF PRESSURE

- Thrust of a gas turbine engine is directly proportional to pressure.



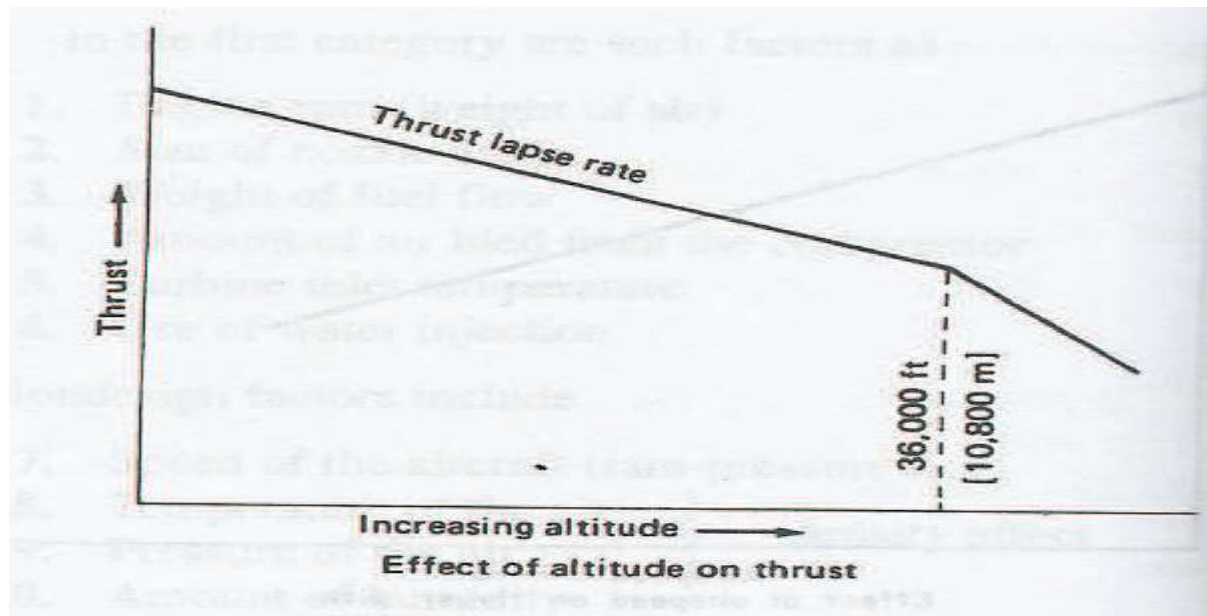
EFFECT OF TEMPERATURE

- As the temperature increases the thrust of the engine reduces and vice-versa.



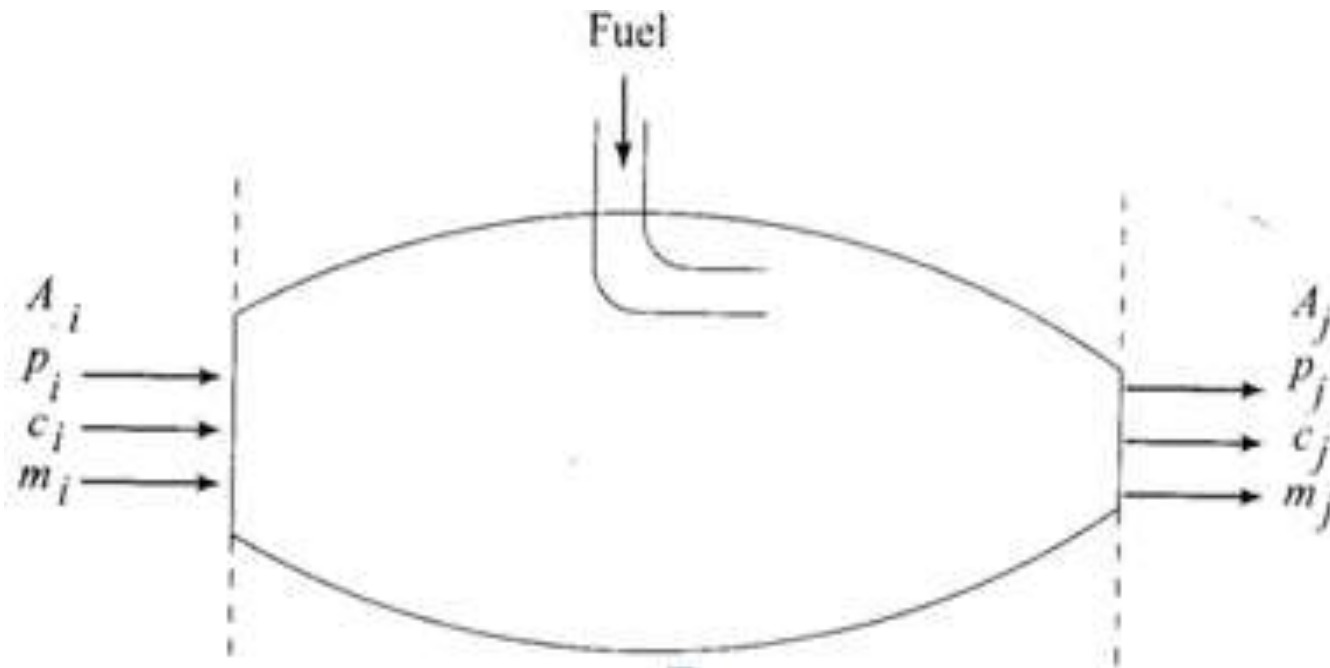
EFFECT OF DENSITY

- With altitude, the pressure and temperature decreases
- Decrease in pressure reduces the thrust and decrease in temperature increases thrust.
- This phenomenon occurs only up to a certain altitude later to which thrust reduces due to reduction in pressure.
- And this reduction in thrust will not be compensated by reduction in temperature.



THRUST EQUATION

- Consider a control volume of a propulsive device
- Area A_i and A_j -large cross sectional at the entry and exit
- m_i -mass of Inlet air through A_i with velocity V_i and Pressure P_i
- M_j -products of combustion exits through the area A_j with velocity V_j and Pressure P_j



- **Total Thrust= Momentum Thrust + Pressure Thrust.....(1)**

$$F_{Total} = F_{momentum} + F_{Pressure}$$

$$F_{moment} = m_j v_j - m_i v_i$$

$$F_{Pressure} = (P_j - P_i)A_j$$

$$F_{Total} = m_j v_j - m_i v_i + (P_j - P_i)A_j$$

m_i =Mass flow rate of air through the engine inlet in Kg/s

m_j = Mass flow rate of exhaust gases through the engine outlet in Kg/s

$$m_j = m_i + m_f$$

m_f =Mass of fuel supplied to the engine in Kg/s

$$m_j = m_i + m_f$$

$$m_j = m_i \left(1 + \frac{m_f}{m_i}\right)$$

$$m_j = m_i (1 + f)$$

$$f = \frac{m_f}{m_i}$$

$$F = m_i (1 + f) v_j - m_i v_i + (P_j - P_i) A_j$$

- Pressure thrust developed will always be small compared to the momentum thrust.
- Example
- The bristol olympus mk. 101
- Momentum thrust of 4216 N and a pressure thrust of 816 N. Giving a total of 5032 kN.

$$F = m_i (1 + f) v_j - m_i v_i$$

- $m_f \ll m_i$

$$F = m_i v_j - m_i v_i$$

$$F = m_i (v_j - v_i)$$

$$F = m_i v_i \left(\frac{v_j}{v_i} - 1 \right)$$

$$F = m_i v_i \left(\frac{1}{\frac{v_i}{v_j}} - 1 \right)$$

$$F = m_i v_i \left(\frac{1}{\alpha} - 1 \right)$$

- **α = Effective Speed Ratio**

$$\alpha = \frac{v_i}{v_j}$$

Thrust Power

- The product of thrust and forward velocity is called the thrust power. It is the power developed by the aircraft in order to overcome drag

$$P_T = F \times v_i$$
$$P_T = m_i(v_j - v_i) \times v_i$$

Specific Thrust F_s

It is the amount of thrust generated per unit flow rate of air.

$$F_s = \frac{F}{\dot{m}_a}$$
$$F_s = \frac{\dot{m}_a(V_2 - V_1)}{\dot{m}_a}$$
$$F_s = (V_2 - V_1)$$

- **In terms of effective speed ratio**

$$F_s = V_2 \left(1 - \frac{V_1}{V_2}\right)$$

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$$F_s = V_2 \left(1 - \frac{V_1}{V_2}\right)$$

$$F_s = V_2 (1 - \alpha)$$

$$\alpha = \frac{V_1}{V_2}$$

- For maximum specific thrust or specific impulse, **APLHA** should be small. If α should be small then V_2 should be large as compared to V_1 .
- On the other hand, if the exhaust velocity V_2 is constant, with forward speed V_1 increases resulting in a reduced specific thrust I_{sp} .
- Hence specific thrust I_{sp} will be maximum when V_1 is zero i.e., $V_1=0$. $I_{sp}=\text{maximum}$.
- The magnitude of specific thrust I_{sp} at $V_1=0$ is of particular significance, because it is a measure of the thrust available at the take-off.

$$F_s = (V_2 - V_1)$$

$$V_1 = 0$$

$$(F_S)_{Static} = (V_2 - 0)$$

- In order to develop large thrust at take-off, the exhaust velocity should be as large as possible.

$$(F_S)_{Static} = V_2$$

- **Specific Impulse I_{SP}**
- It is defined as the thrust developed per unit weight flow rate of air through the propulsive device.

$$I_{SP} = \frac{T}{\dot{W}_a}$$

- **Specific Fuel Consumption SFC**
- **Defined as the rate of fuel consumption per unit thrust**
- **For Jet engines SFC- TSFC is used**

$$TSFC = \frac{\dot{m}_f}{T}$$

$$TSFC = \frac{\dot{m}_f}{T} * 3600 \text{ kg} / \text{N} - \text{thrust} - \text{hour}$$

Thermal Efficiency of a Turbojet Engine

- It is defined as the ratio of the propulsive power to the heat supplied

$$\eta_{Th} = \frac{\text{Propulsive- Power}}{\text{Heat- Supplied}}$$

- Propulsive power is the difference between the kinetic energy of the working fluid between the outlet and the inlet.

$$\text{Propulsive- Power} = \frac{1}{2}(\dot{m}_a + \dot{m}_f)V_2^2 - \frac{1}{2}\dot{m}_aV_1^2$$

$$\dot{m}_a + \dot{m}_f = \dot{m}_a$$

$$\text{Propulsive- Power} = \frac{1}{2}\dot{m}_aV_2^2 - \frac{1}{2}\dot{m}_aV_1^2$$

$$\text{Propulsive- Power} = \frac{\dot{m}_a}{2}(V_2^2 - V_1^2)$$

- In terms of effective speed ratio

$$\text{Propulsive- Power} = \frac{\dot{m}_a}{2} V_2^2 \left(1 - \frac{V_1^2}{V_2^2} \right)$$

$$\text{Propulsive- Power} = \frac{\dot{m}_a}{2} V_2^2 (1 - \alpha^2)$$

- Thermal efficiency is

$$\eta_{Th} = \frac{\frac{\dot{m}_a}{2} V_2^2 (1 - \alpha^2)}{\dot{m}_f * C_v}$$

$$\eta_{Th} = \frac{\dot{m}_a V_2^2 (1 - \alpha^2)}{2 \dot{m}_f * C_v}$$

$$f = \frac{\dot{m}_f}{\dot{m}_a}$$

$$\eta_{Th} = \frac{V_2^2 (1 - \alpha^2)}{2 f * C_v}$$

Propulsive Efficiency

- It is defined as the ratio of the thrust power to the propulsive power.

$$\eta_p = \frac{\text{Thrust- Power}}{\text{Propulsive- Power}}$$

$$\eta_p = \frac{F * V_1}{\frac{1}{2} \dot{m}_a (V_2^2 - V_1^2)}$$

$$\eta_p = \frac{\dot{m}_a (V_2 - V_1) * V_1}{\frac{1}{2} \dot{m}_a (V_2^2 - V_1^2)}$$

$$\eta_p = \frac{2(V_2 - V_1)V_1}{(V_2 - V_1)(V_2 + V_1)}$$

$$\eta_p = \frac{2V_1}{(V_2 + V_1)}$$

- In terms of effective speed ratio

$$\eta_p = \frac{2}{\frac{(V_2 + V_1)}{V_1}}$$

$$\eta_p = \frac{2}{\frac{V_2}{V_1} + 1}$$

$$\eta_p = \frac{2}{\frac{1}{\alpha} + 1}$$

$$\eta_p = \frac{2}{\frac{1 + \alpha}{\alpha}}$$

$$\eta_p = \frac{2\alpha}{1 + \alpha}$$

Overall Efficiency

- It is defined as the ratio of the thrust power to the heat supplied.

$$\eta_o = \frac{\text{Thrust} - \text{Power}}{\text{Heat} - \text{Supplied}}$$

$$\eta_o = \frac{\dot{m}_a (V_2 - V_1) * V_1}{\dot{m}_f * C_v}$$

- Multiply and Divide by Propulsive Power**

$$\text{Propulsive} - \text{Power} = \frac{\dot{m}_a}{2} (V_2^2 - V_1^2)$$

$$\eta_o = \frac{\dot{m}_a (V_2 - V_1) * V_1}{\frac{\dot{m}_a}{2} (V_2^2 - V_1^2)} * \frac{\frac{\dot{m}_a}{2} (V_2^2 - V_1^2)}{\dot{m}_f * C_v}$$

$$\eta_o = \eta_P * \eta_{Th}$$

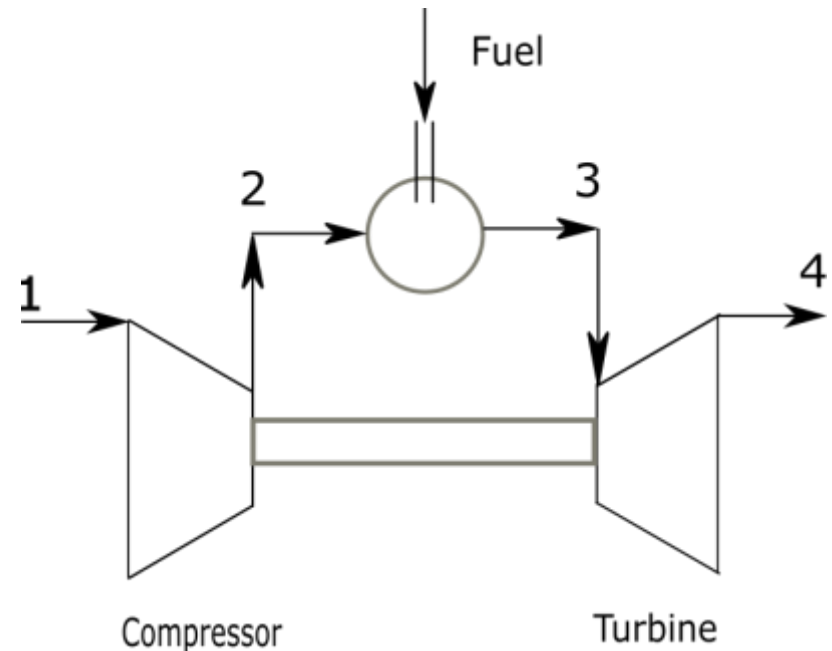
Performance of an Ideal Brayton Cycle

- **Parameters**

- Specific Power Output
- Cycle Efficiency

- **Assumptions**

- Compression & Expansion-Isentropic
- Change in KE of the working fluid is negligible between inlet and outlet of each component
- No pressure losses in any component
- Working fluid-Perfect gas
- Mass flow is constant
- No frictional losses



- Consider steady flow energy equation

$$Q + h_1 + gz_1 + \frac{1}{2}V_1^2 = Ws + h_2 + gz_2 + \frac{1}{2}V_2$$

- In most of the turbo machineries like the compressor and the turbine are considered as adiabatic machines i.e., $Q=0$.
- Further, the changes in potential energy is also negligible i.e., $Z_1-Z_2 = 0$
- Using these conditions, we get the SFEE applied for a gas turbine engine to be

$$Q + h_1 + gz_1 + \frac{1}{2}V_1^2 = Ws + h_2 + gz_2 + \frac{1}{2}V_2$$

$$h_1 + \frac{1}{2}V_1^2 = Ws + h_2 + \frac{1}{2}V_2$$

- Also as per the assumption, KE is zero

$$\frac{1}{2}V_1^2 = \frac{1}{2}V_2 = 0$$

$$h_1 = Ws + h_2$$

- We get the shaft work to be

$$Ws = h_1 - h_2$$

$$Ws = -(h_2 - h_1)$$

- W_s = work transfer per unit mass flow rate
- h = enthalpy
- Apply this equation to each component of the gas turbine engine

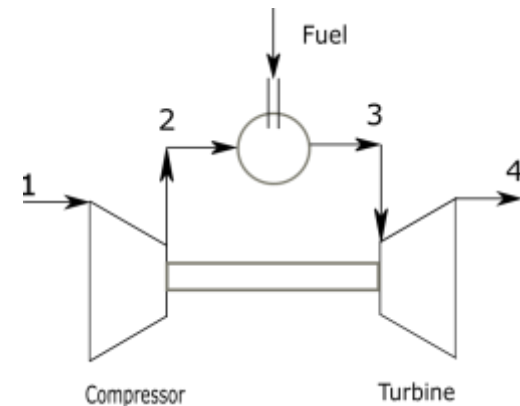
Specific Power Output

- **Compressor Work (W_c)** $Ws = (h_2 - h_1)$

$$Ws = C_p (T_2 - T_1)$$

- **Heat Added Q** $Q = (h_3 - h_2)$
 $Q = C_p (T_3 - T_2)$

- **Turbine Work (W_T)** $W_T = (h_3 - h_4)$
 $W_T = C_p (T_3 - T_4)$



- **Net Work Done W_N**

$$W_N = W_T - W_C$$

$$W_N = W_{34} - W_{12}$$

$$W_N = C_P(T_3 - T_4) - C_P(T_2 - T_1)$$

$$W_N = C_P(T_3 - T_4 - T_2 + T_1)$$

$$W_N = C_P T_1 \left(\frac{T_3}{T_1} - \frac{T_4}{T_1} - \frac{T_2}{T_1} + \frac{T_1}{T_1} \right)$$

$$W_N = C_P T_1 \left(\frac{T_3}{T_1} - \frac{T_4}{T_1} - \frac{T_2}{T_1} + 1 \right)$$

- Take $\frac{T_3}{T_1} = t$
- From isentropic relations

- $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\left(\frac{\gamma-1}{\gamma} \right)}$$

- For an ideal cycle, the pressure ratio is

$$\frac{P_2}{P_1} = \frac{P_3}{P_4} = r$$

- Thus

$$\frac{T_2}{T_1} = r^{\left(\frac{\gamma-1}{\gamma} \right)} = \frac{T_3}{T_4}$$

- .

$$W_N = C_p T_1 \left(\frac{T_3}{T_1} - \frac{T_4}{T_3} * \frac{T_3}{T_1} - \frac{T_2}{T_1} + 1 \right)$$

$$\frac{W_N}{C_p T_1} = \left(t - \frac{1}{r^{\left(\frac{\gamma-1}{\gamma}\right)}} * t - r^{\left(\frac{\gamma-1}{\gamma}\right)} + 1 \right)$$

$$\frac{W_N}{C_p T_1} = t \left(1 - \frac{1}{r^{\left(\frac{\gamma-1}{\gamma}\right)}} \right) - \left(r^{\left(\frac{\gamma-1}{\gamma}\right)} + 1 \right)$$

- Replace $r^{\left(\frac{\gamma-1}{\gamma}\right)} = c$

$$\frac{W_N}{C_p T_1} = t \left(1 - \frac{1}{c} \right) - (c + 1)$$

- This equation shows that the specific work output of a gas turbine engine is dependent on both the maximum pressure ratio of the cycle as well as the maximum cycle temperature .

- **Cycle Efficiency**

$$\eta = \frac{\text{Net - Work - Output}}{\text{Heat - Supplied}}$$

$$\eta = \frac{C_p T_1 \left(t \left(1 - \frac{1}{c} \right) - (c + 1) \right)}{C_p (T_3 - T_2)}$$

$$\eta = \frac{C_p T_1 \left(t \left(1 - \frac{1}{c} \right) - (c + 1) \right)}{C_p T_1 \left(\frac{T_3}{T_1} - \frac{T_2}{T_1} \right)}$$

$$\eta = \frac{t \left(1 - \frac{1}{c} \right) - (c + 1)}{t - c}$$

$$\eta = \frac{t - \frac{t}{c} - c + 1}{t - c}$$

$$\eta = \frac{(t - c) - \left(\frac{t}{c} - 1 \right)}{t - c}$$

$$\eta = \frac{(t-c) - \left(\frac{t-c}{c}\right)}{t-c}$$

$$\eta = \frac{(t-c) \left(1 - \frac{1}{c}\right)}{(t-c)}$$

$$\eta = 1 - \frac{1}{c}$$

$$\eta = 1 - \frac{1}{r^{\left(\frac{\gamma-1}{\gamma}\right)}}$$

- Cycle efficiency of a gas turbine engine depends only on the maximum pressure ratio

Numericals

1) A turbojet power plant uses aviation kerosene fuel having a calorific value of 43MJ/kg. The fuel consumption is 0.18kg per hour per Newton of thrust, when the thrust is 9kN. The aircraft velocity is 500m/s and the mass of air passing through the compressor is 27kg/s. Calculate the air-fuel ratio and the overall efficiency.

Solution

- Calorific Value of the Fuel $CV=43\text{MJ/kg}$
- Fuel Consumption Rate $\dot{m}_f = 0.18\text{kg} / \text{h} / \text{N}$
- Thrust of the engine $T=9000\text{N}$
- Forward Velocity of the aircraft $V_1=500\text{m/s}$

Mass flow rate of fuel

$$\dot{m}_f = 0.18\text{kg} / \text{h} / \text{N}$$

$$\dot{m}_f = \frac{0.18}{3600} * 9000$$

$$\dot{m}_f = 0.45\text{kg} / \text{s}$$

- **Air-Fuel ratio**

$$f = \frac{\dot{m}_a}{\dot{m}_f}$$

$$f = \frac{27}{0.45}$$

$$f = 60:1$$

- **Thrust Power P_T**

$$P_T = F * V_1$$

$$P_T = 9000 * 500$$

$$P_T = 4500kW$$

- **Heat Supplied to the Engine Q**

$$Q = \dot{m}_f * CV$$

$$Q = 0.45 * 43000$$

$$Q = 19350kW$$

- **Overall Efficiency**

$$\eta = \frac{P_T}{Q}$$

$$\eta = \frac{4500}{19350}$$

$$\eta = 0.2326$$

$$\eta = 23.26\%$$

2) The effective jet exit velocity from a jet engine is 2700m/s. The forward flight velocity is 1350m/s and the air flow rate is 78.6kg/s. Calculate the following:

1. Thrust
2. Thrust Power
3. Propulsive Efficiency

- Forward Velocity of the flight $V_1=1350\text{m/s}$
- Exit velocity of the gases $V_2=2700\text{m/s}$
- Mass flow rate of air

Solution

- Effective jet velocity

$$\alpha = \frac{V_1}{V_2}$$

$$\alpha = \frac{1350}{2700}$$

$$\alpha = 0.5$$

- **Thrust**

$$F = \dot{m}_a (V_2 - V_1)$$

$$F = 78.6 * (2700 - 1350)$$

$$F = 106110N$$

- **Thrust Power**

$$P_T = F * V_1$$

$$P_T = 16110 * 1350$$

$$P_T = 143.25 * 10^6 W$$

- **Propulsive Power**

$$\eta_p = \frac{2\alpha}{1 + \alpha}$$

$$\eta_p = \frac{2 * 0.5}{1 + 0.5}$$

$$\eta_p = 0.6667$$

$$\eta_p = 66.67\%$$

- 3) The diameter of the propeller of an aircraft is 2.5m. It flies at a speed of 500kmph at an altitude of 8000m. For a flight to jet speed ratio of 0.75, determine
 1. Flow rate of air through the propeller
 2. Thrust
 3. Specific thrust
 4. Specific impulse
 5. Thrust power

At $h=8000\text{m}$, take $\rho=0.525\text{kg/m}^3$

Solution

- Cross sectional area of the propeller

$$A = \frac{\pi d^2}{4}$$

$$A = \frac{\pi * 2.5^2}{4}$$

$$A = 4.908\text{m}^2$$

- **Flight Speed**

$$V_1 = 500 \text{ kmph}$$

$$V_1 = \frac{500 * 1000}{3600}$$

$$V_1 = 138.9 \text{ m/s}$$

- **Effective Jet ratio**

$$\alpha = \frac{V_1}{V_2}$$

$$0.75 = \frac{138.98}{V_2}$$

$$V_2 = 185.18 \text{ m/s}$$

Flow rate of air through the propeller

- Velocity of air through the propeller

$$V = \frac{1}{2}(V_1 + V_2)$$

$$V = 0.5 * (138.98 + 185.18)$$

$$V = 162.035 \text{ m/s}$$

- Mass flow rate through the propeller

$$\dot{m}_a = \rho AV$$

$$\dot{m}_a = 0.525 * 4.908 * 162.035$$

$$\dot{m}_a = 417.516 \text{ kg/s}$$

- Thrust

$$F = \dot{m}_a (V_2 - V_1)$$

$$F = 417.516 * (185.16 - 138.89)$$

$$F = 19.3268 \text{ kN}$$

- **Specific Thrust**

$$F_s = \frac{F}{\dot{m}_a}$$

$$F_s = \frac{19.3268}{417.516}$$

$$F_s = 46.29N / (kg / s)$$

- **Specific Impulse I_{SP}**

$$I_{SP} = \frac{F}{\dot{w}_a}$$

$$I_{SP} = \frac{F}{\dot{m}_a * g}$$

$$I_{SP} = \frac{46.29}{9.81}$$

$$I_{SP} = 4.718 \text{Seconds}$$

- **Thrust Power P_T**

$$P_T = F * V_1$$

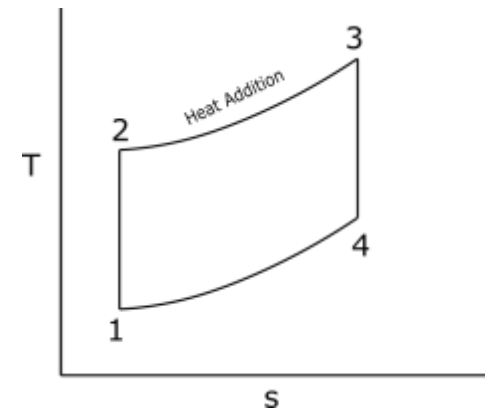
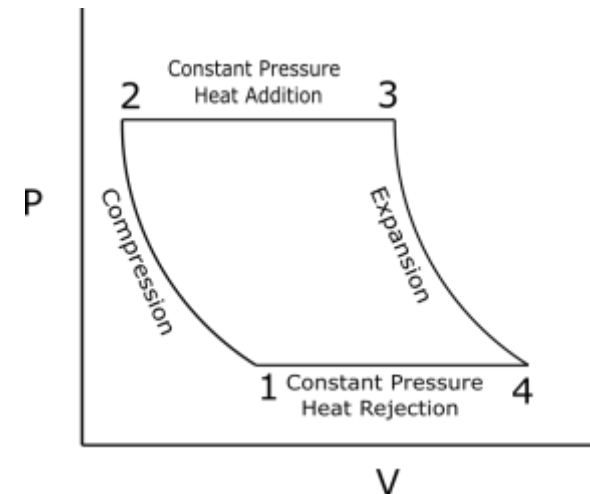
$$P_T = 19.3268 * 138.89$$

$$P_T = 2684.3kW$$

- 4) In an ideal Brayton cycle, air from atmosphere at 1bar and 300K is compressed to 6bars and the maximum cycle temperature is limited to 1100⁰K. If the heat supplied is 100MW, the find
- Thermal Efficiency of the cycle
- Net work done
- Work ratio

Solution

- Pressure of air at inlet $P_1 = 1$ Bar
- Temperature $T_1 = 300$ K
- Pressure of air after compression $P_2 = 6$ Bar
- Maximum cycle temperature $T_3 = 1100$ K
- Heat supplied $Q = 100$ MW



- **Thermal Efficiency of the cycle**

$$\eta = 1 - \frac{1}{(r)^{\frac{\gamma-1}{\gamma}}}$$

$$\eta = 1 - \frac{1}{(6)^{\frac{1.4-1}{1.4}}}$$

$$\eta = 40\%$$

- **Net work done**

$$W_C = C_p(T_2 - T_1)$$

$$W_T = C_p(T_3 - T_4)$$

- We need T_2 and T_4 for calculation of work output

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\left(\frac{\gamma-1}{\gamma} \right)}$$

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\left(\frac{\gamma-1}{\gamma} \right)}$$

$$T_2 = T_1 (r)^{\left(\frac{\gamma-1}{\gamma} \right)}$$

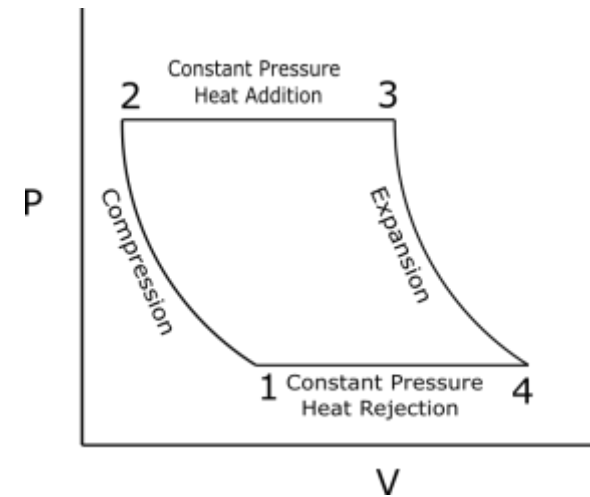
$$T_2 = 300 * (6)^{\left(\frac{1.4-1}{1.4} \right)}$$

$$T_2 = 501K$$

- In an ideal cycle

$$r_{Compr} = r_{turbine}$$

- So $P_2 = P_3$ and $P_1 = P_4$ on the P-V diagram



$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4} \right)^{\left(\frac{\gamma-1}{\gamma} \right)} = (r_{turbine})$$

$$\frac{T_3}{T_4} = \left(\frac{6}{1} \right)^{\left(\frac{1.4-1}{1.4} \right)}$$

$$\frac{T_3}{T_4} = 1.67$$

$$T_4 = \frac{T_3}{1.67}$$

$$T_4 = \frac{1100}{1.67}$$

$$T_4 = 658.4K$$

- Net Work Done

$$W_N = W_T - W_C$$

$$W_C = C_p (T_2 - T_1)$$

$$W_C = 1.005(501 - 300)$$

$$W_C = 202 \text{ kJ / kg}$$

$$W_T = C_p (T_3 - T_4)$$

$$W_T = 1.147(1100 - 658.7)$$

$$W_T = 506.17 \text{ kJ / kg}$$

$$W_N = W_T - W_C$$

$$W_N = 506.17 - 202$$

$$W_N = 304.17 \text{ kJ / kg}$$

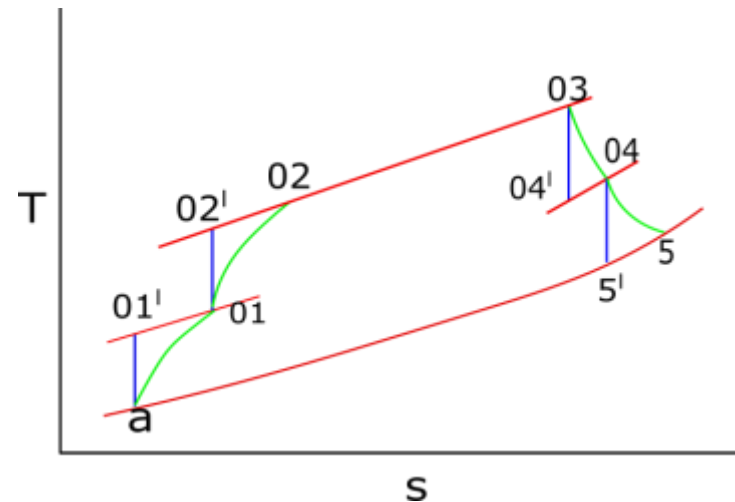
- Work Ratio

$$WR = \frac{506.17 - 202}{506.17}$$

$$WR = 0.6$$

- 5) A turboprop aircraft is flying at 600kmph at an altitude where the ambient conditions are 0.458 bar and -15 degree Celsius. Compressor pressure ratio is 9:1 and max gas temperature is 1200K. The intake duct efficiency is 0.9 and total head isentropic efficiency of compressor and turbine are 0.89 and 0.93 respectively. Calculate the specific power output in kJ/kg, thermal efficiency taking the mechanical efficiency of transmission as 98% and neglecting the losses other than specified. Assume that the exhaust gases leave the aircraft at 600kmph relative to the aircraft.

- $V_1 = 600 \text{ kmph} = 166.67 \text{ m/s}$
- $P_a = 0.458 \text{ bar}$
- $T_a = -15^\circ\text{C} = 258 \text{ K}$
- Compressor pressure ratio $r_c = 9:1$
- $T_{03} = 1200 \text{ K}$
- $V_2 = 600 \text{ kmph} = 166.67 \text{ m/s}$
- Diffuser efficiency $\eta_{\text{Diff}} = 0.9$
- Compressor Efficiency $\eta_{\text{Comp}} = 0.89$
- Turbine Efficiency $\eta_{\text{Turb}} = 0.93$
- Transmission Efficiency $\eta_{\text{Trnsm}} = 0.98$



$$T_{01}' = T_a + \frac{V_1^2}{2C_p}$$

$$T_{01}' = 258 + \frac{166.67^2}{2 * 1005}$$

$$T_{01}' = 271.82K = T_{01}$$

$$\frac{P_{01}'}{P_a} = \left(\frac{T_{01}'}{T_a} \right)^{\frac{\gamma}{\gamma-1}}$$

$$P_{01}' = P_a * \left(\frac{T_{01}'}{T_a} \right)^{\frac{\gamma}{\gamma-1}}$$

$$P_{01}' = 0.458 * \left(\frac{271.82}{258} \right)^{\frac{1.4}{1.4-1}}$$

$$P_{01}' = 0.5497Bar$$

$$\eta_{Ram} = \frac{P_{01} - P_a}{P_{01}' - P_a}$$

$$\eta_{Ram} = \frac{P_{01} - P_a}{P_{01}' - P_a}$$

$$P_{01} = \eta_{Ram} (P_{01}' - P_a) + P_a$$

$$P_{01} = \eta_{Ram} (P_{01}' - P_a) + P_a$$

$$P_{01} = 0.5405bar$$

$$\frac{P_{02}}{P_{01}} = r_c$$

$$P_{02} = r_c * P_{01}$$

$$P_{02} = 4.865bar$$

$$\frac{T_{02}'}{T_{01}} = r^{\frac{\gamma-1}{\gamma}}$$

$$T_{02}' = T_{01} * r^{\frac{\gamma-1}{\gamma}}$$

Because $T_{01}' = T_{01}$

$$T_{02}' = 271.85 * 9^{\frac{\gamma-1}{\gamma}}$$

$$T_{02}' = 509.6K$$

Compressor Efficiency

$$\eta_c = \frac{T_{02}' - T_{01}}{T_{02} - T_{01}}$$

$$T_{02} = \frac{T_{02}' - T_{01}}{\eta_C} + T_{01}$$

$$T_{02}=\frac{509.60-271.82}{0.89}+271.82$$

$$T_{02}=539K$$

$$\frac{T_{o4}}{T_{03}}^{' }=\left(\frac{P_{04}}{P_{03}}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_{04}}^{' }=T_{03}}*\left(\frac{P_{04}}{P_{03}}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_{04}}^{' }=1200*\left(\frac{1}{9}\right)^{0.248}$$

$$T_{04}}^{' }=695.87K$$

$$\eta_T=\frac{T_{03}-T_{04}}{T_{03}-T_{04}}^{' },$$

$$T_{03}-T_{04}=\eta_T*\left(T_{03}-T_{04}}^{' }\right)$$

$$T_{04} = T_{03} - \eta_T * (T_{03} - T_{04}')$$

$$T_{04} = 1200 - 0.93 * (1200 - 695.87)$$

$$T_{04} = 731.16K$$

$$W_N = W_T - W_C$$

$$W_N = W_T - \frac{1}{\eta_{Tr}} * W_C$$

$$W_N = C_{pg} (T_{03} - T_{04}) - \frac{1}{\eta_{Tr}} * C_{pa} (T_{02} - T_{01})$$

$$W_N = 1.147 * (1200 - 731.16) - \frac{1}{0.98} * 1.005 * (539 - 271.82)$$

$$W_N = 236.kJ/kg$$

$$\eta_{TH} = \frac{W_N}{Q}$$

$$\eta_{TH} = \frac{W_N}{C_{p_g}(T_{03} - T_{04})}$$

$$\eta_{TH} = \frac{236}{1.147*(1200-539)}$$

$$\eta_{TH} = 34.79\%$$